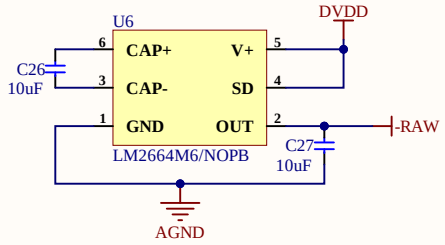
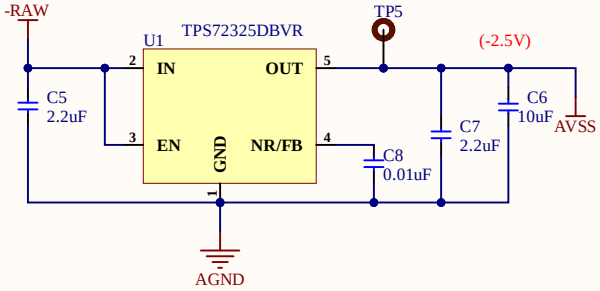
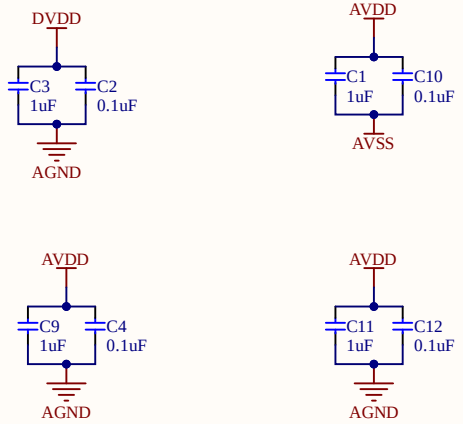
## VOLTAGE_INVERTER




## -2.5V_REGULATOR



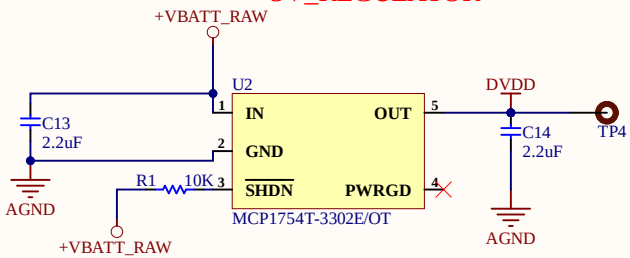

## ADS_DECOUPLING



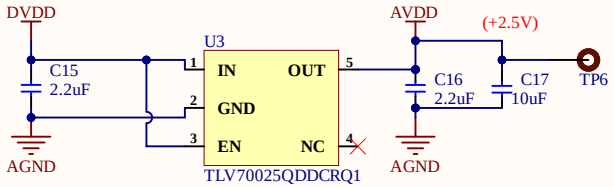


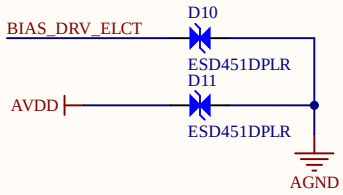


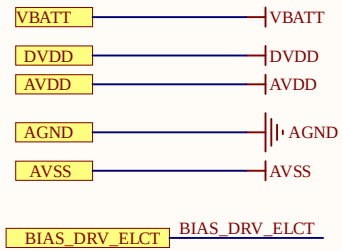
## 3V_REGULATOR




## +2.5V_REGULATOR



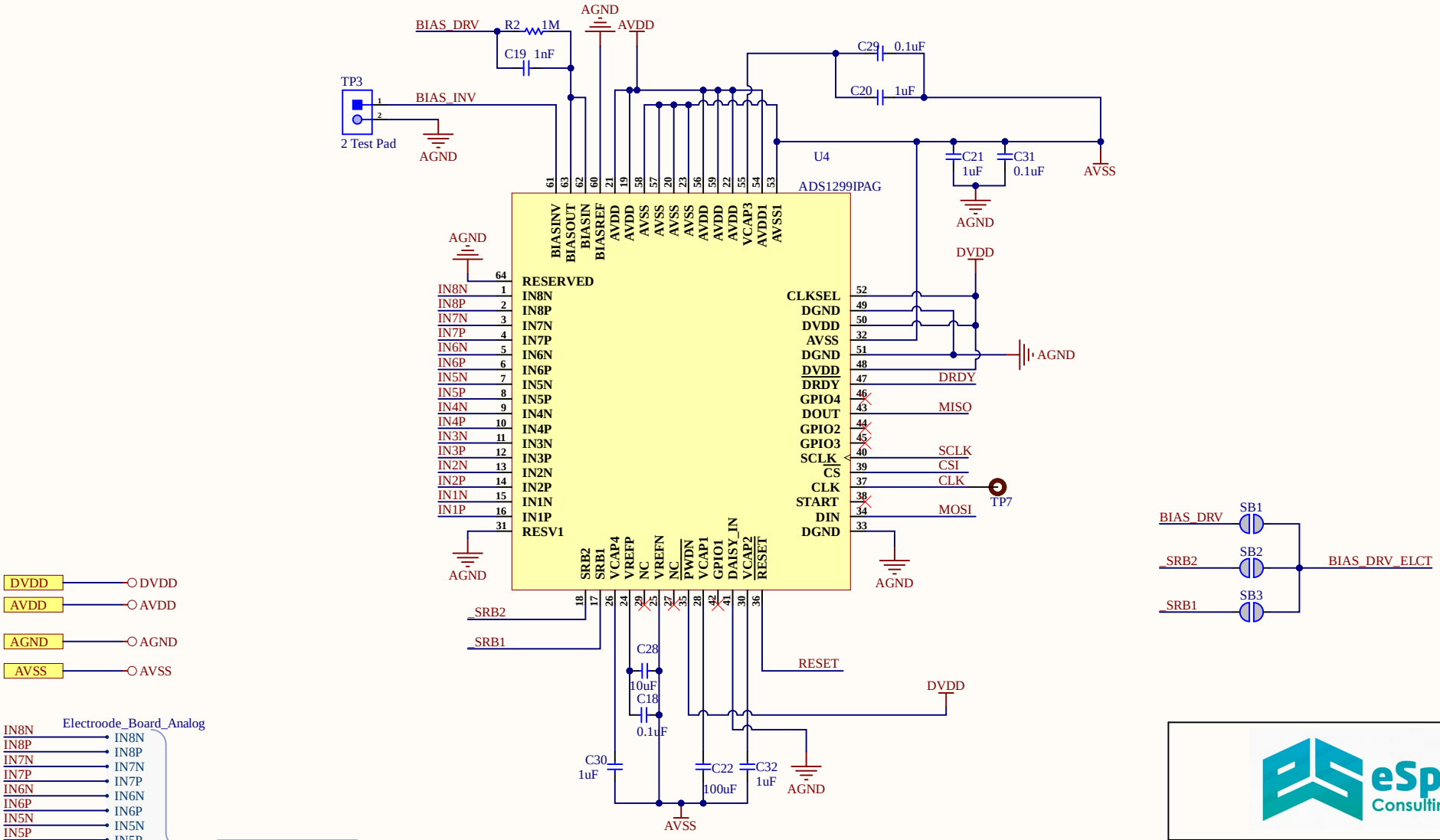




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| Title: Power_Sheet.SchDoc | |
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| Date: 10/02/2025 | Page Num: 1 |
| Drawn By: Huzaifa Ali | Page Total: 4 |

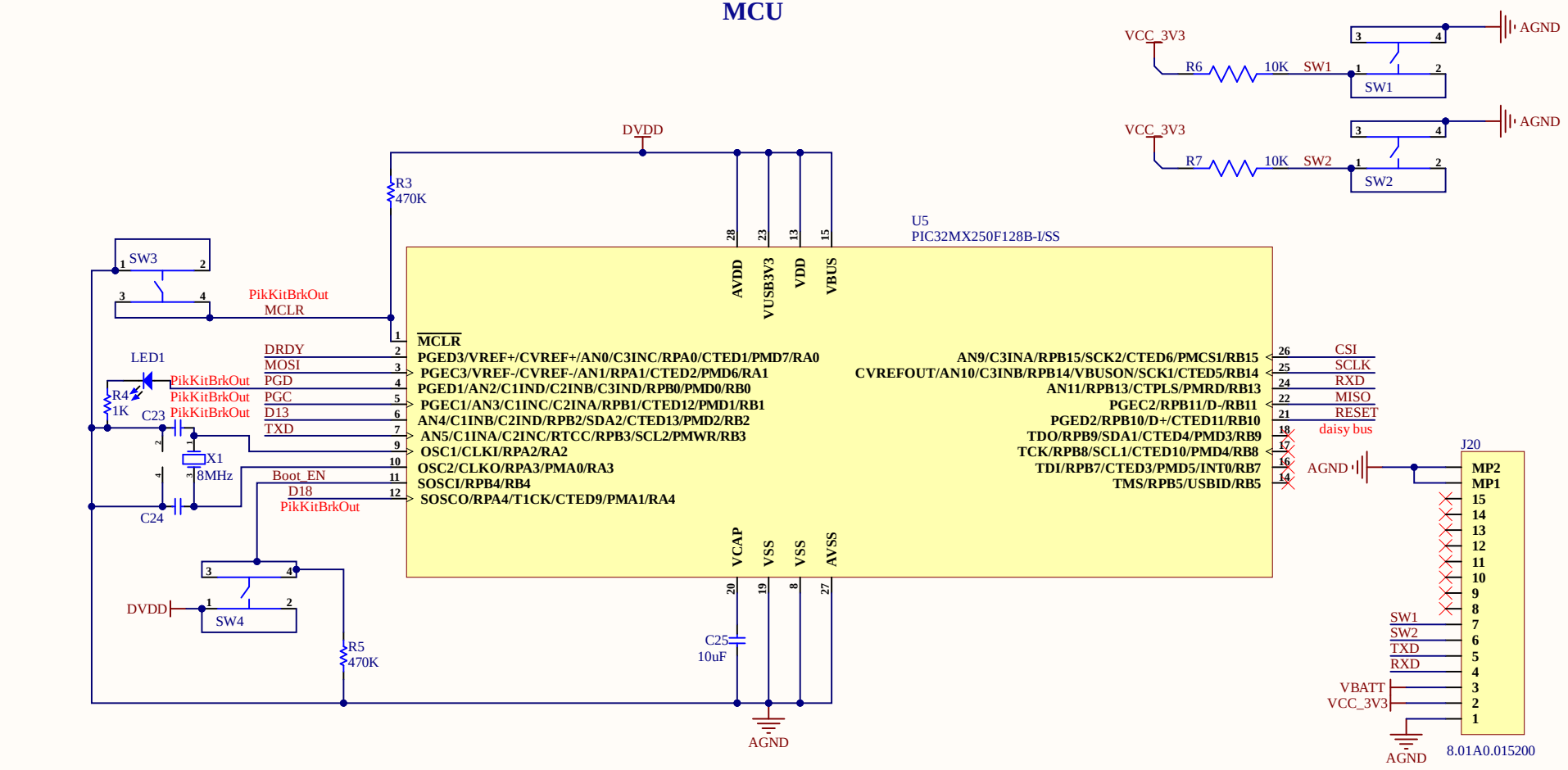
# ANALOG FRONT END (AFE)

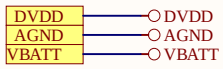


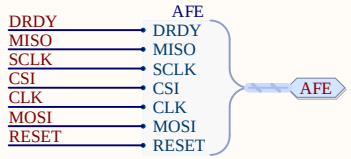

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| Date: 10/02/2025 | Page Num: 2 |
| Drawn By: Huzaifa Ali | Page Total: 4 |

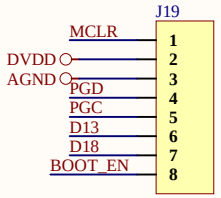
## SENSOR OUT SIGNALS

## MCU



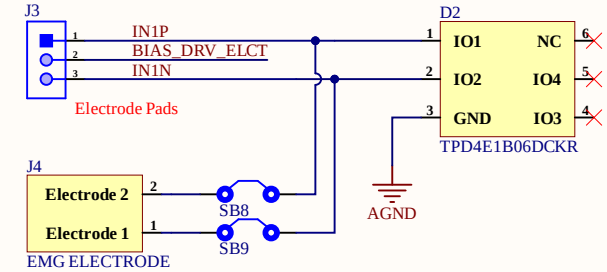



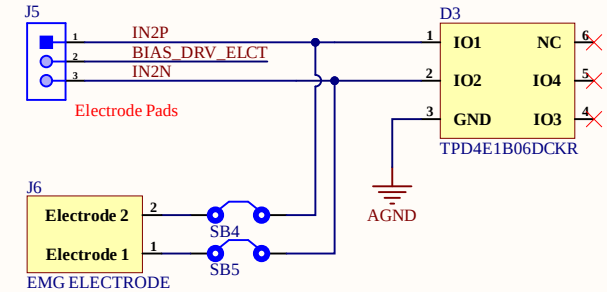



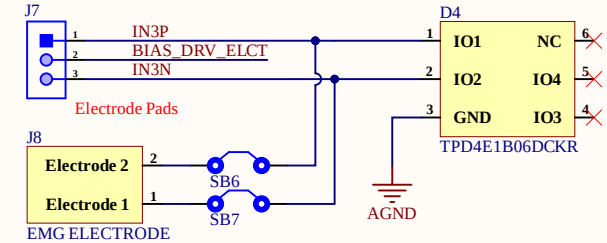



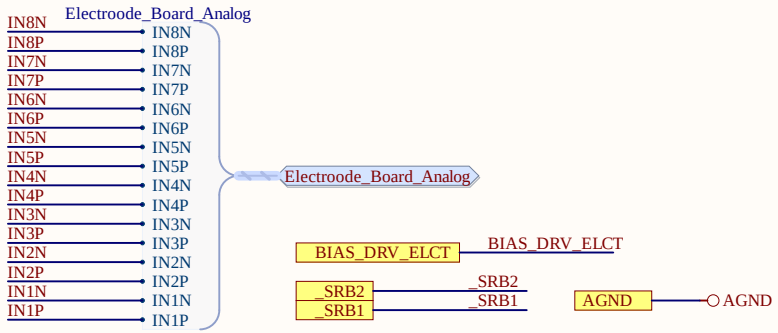

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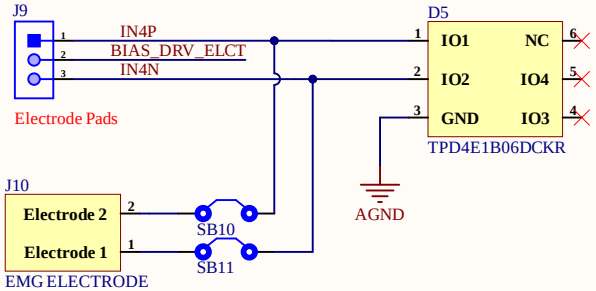
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| Title: Pic_Sheet.SchDoc | | |
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| Date: 11/02/2025 | Page | Num: 3 |
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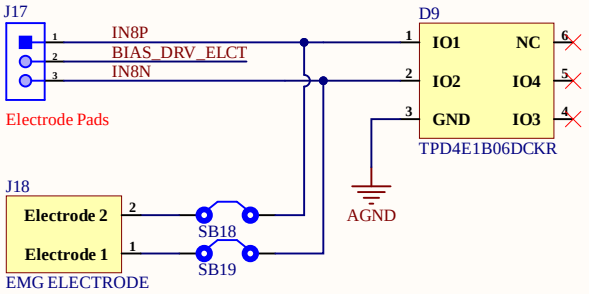


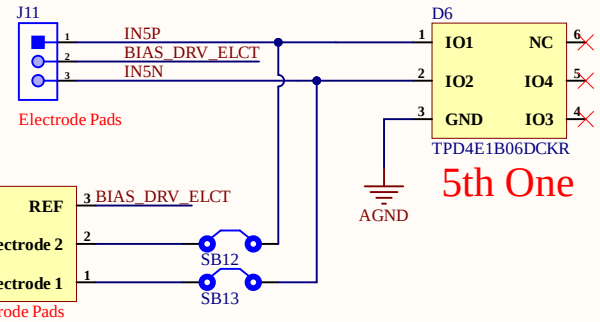



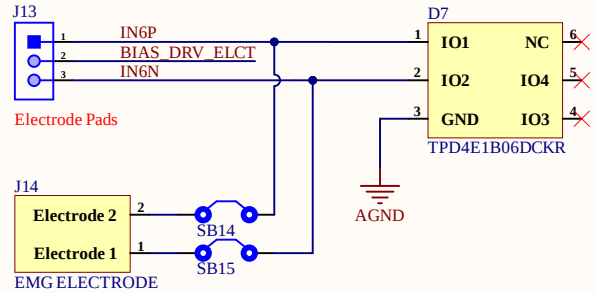



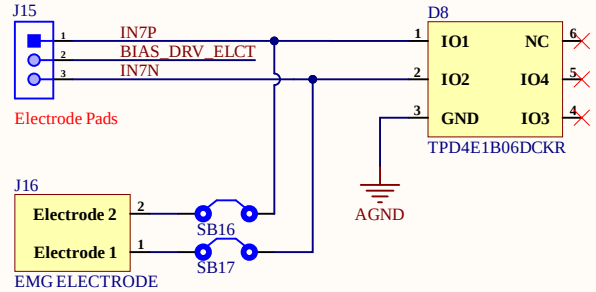



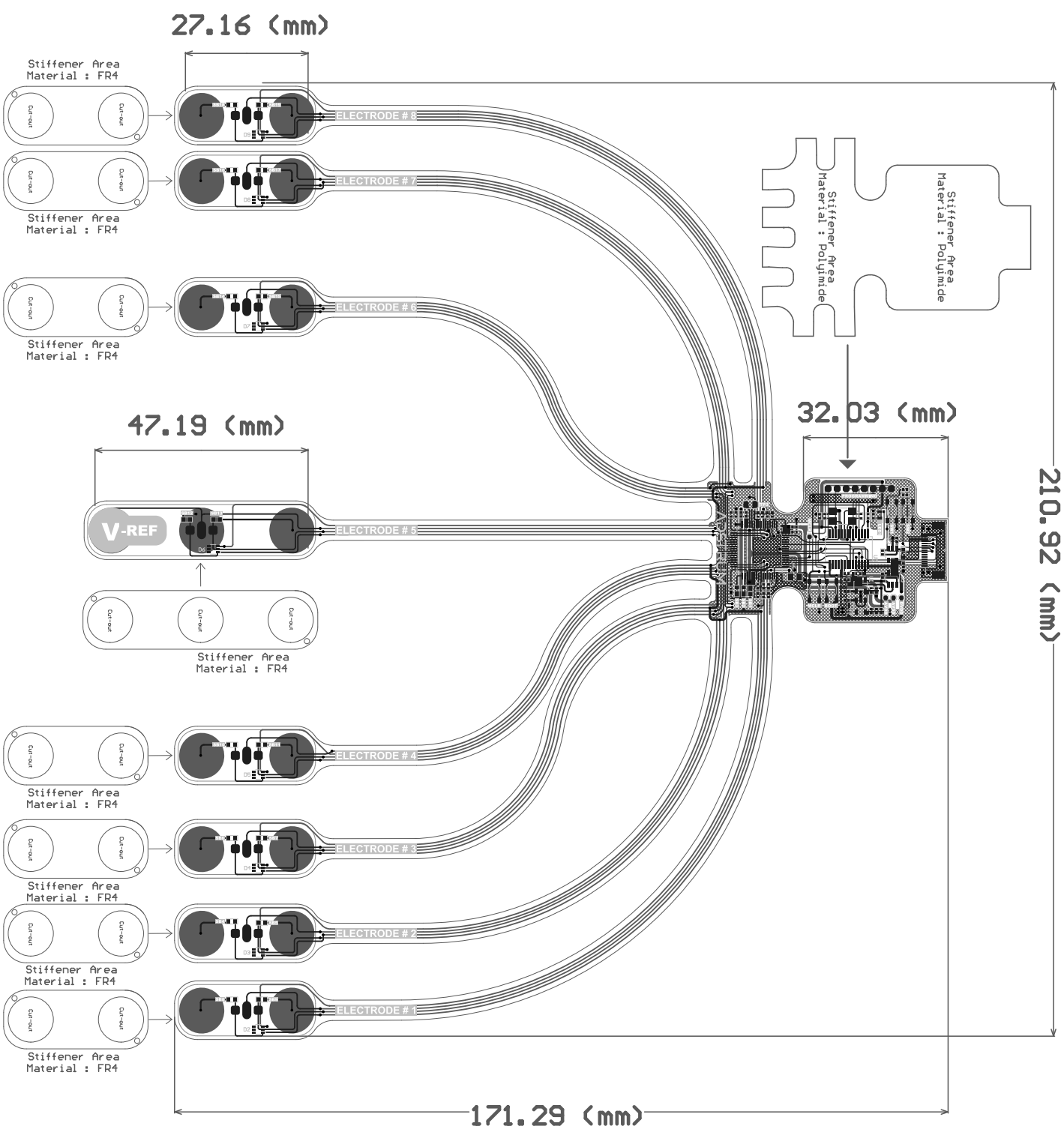




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|---|---|---|
| Title: Electrode_Sheet.SchDoc | | |
| File: EMG_Arm_ELCT-V3.0.PrjPcb | | |
| Date: 10/02/2025 | Page | Num: 4 |
| Drawn By: Huzaifa Ali | Page | Total: 4 |




| Layer | Name | Material | Thickness | Constant | Flex Layer Stack |
|---|---|---|---|---|---|
| | Top Overlay | | | | |
| | Top Solder | Solder Resist | 0.013mm | 3.5 | |
| 1 | Top Layer | | 0.035mm | | |
| | Dielectric 1 | Flex | 0.020mm | 4.8 | |
| 2 | Bottom Layer | | 0.035mm | | |
| | Bottom Solder | Solder Resist | 0.013mm | 3.5 | |
| | Bottom Overlay | | | | |

Total board thickness: 0.116mm

| Comment | Description | Designator | Footprint | LibRef | Quantity |
|---|---|---|---|---|---|
| 1uF | CAP CER 1UF 25V X5R 0402 | C1, C3, C9, C11, C20, C21, C30, C32 | Cap-0402 | CC0402KRX5R8BB105 | 8 |
| 0.1uF | CAP CER 0.1UF 16V X7R 0402 | C2, C4, C10, C12, C18, C29, C31 | Cap-0402 | CC0402KRX7R7BB104 | 7 |
| 2.2uF | CAP CER 2.2UF 35V X5R 0402 | C5, C7, C13, C14, C15, C16 | Cap-0402 | C1005X5R1V225K050BC | 6 |
| 10uF | CAP CER 10UF 10V X5R 0402 | C6, C17, C25, C26, C27, C28 | Cap-0402 | CC0402MRX5R6BB106, CL05A106MP5NUNC | 6 |
| 0.01uF | CAP CER 10000PF 25V X5R 0402 | C8 | Cap-0402 | CC0402KRX5R8BB103 | 1 |
| 1nF | CAP CER 1000PF 25V X7R 0402 | C19 | Cap-0402 | CC0402KRX7R8BB102 | 1 |
| 100uF | CAP CER 100UF 6.3V X5R 0805 | C22 | Cap-0805 | GRM21BR60J107ME15L | 1 |
| 25pF | CAP CER 25PF 50V C0G/NPO 0402 | C23, C24 | Cap-0402 | CC0402JRNPO9BN250 | 2 |
| CDBU0130L | DIODE SCHOTTKY 30V 100MA 0603C | D1 | CDBU0130L-HF | CDBU0130L | 1 |
| TPD4E1B06DCKR | TVS DIODE 5.5VWM 14.5VC SC706 | D2, D3, D4, D5, D6, D7, D8, D9 | SOT-363 | TPD4E1B06DCKR | 8 |
| ESD451DPLR | TVS DIODE 5.5VWM 9.5VC 2X2SON | D10, D11 | X2SON-2(0.3x0.6) | ESD451DPLR | 2 |
| EMG ELECTRODE | Sensor Connecting Pads | J4, J6, J8, J10, J14, J16, J18 | EMG 2 PAD | EMG ELECTRODE | 7 |
| Electrode Pads | Sensor Connecting Pads | J12 | EMG 3 Pad | EMG ELECTRODE 3 Pin | 1 |
| SMD PADS (1x8) 2.54mm | | J19 | SMD PADS (1x8) 2.54mm | SMD PADS (1x8) 2.54mm | 1 |
| 8.01A0.015200 | -40℃~+80℃ 15P Dual-sided contacts / top and bottom connection 0.5mm SMD,P=0.5mm,Surface Mount, Right Angle FFC, FPC (Flat Flexible) Connector Assemblies ROHS | J20 | 8.01A0.015200 | 8.01A0.015200 | 1 |
| 150060BS75000 | LED BLUE CLEAR 0603 SMD | LED1 | LED-0603 | 150060BS75000 | 1 |
| 10K | RES SMD 10K OHM 0.1% 1/16W 0402, RES 10K OHM 1% 1/16W 0402 | R1, R6, R7 | Res-0402 | RT0402BRD0710KL, RC0402FR-0710KL | 3 |
| 1M | RES 1M OHM 1% 1/16W 0402 | R2 | Res-0402 | RC0402FR-071ML | 1 |
| 470K | RES 470K OHM 1% 1/16W 0402 | R3, R5 | Res-0402 | RC0402FR-07470KL | 2 |
| 1K | RES 1K OHM 1% 1/16W 0402 | R4 | Res-0402 | RC0402FR-071KL | 1 |
| PTS810SJK250SMTRLFS | SWITCH TACTILE SPST-NO 0.05A 16V | SW1, SW2, SW3, SW4 | PTS810SJK250SMTRLFS | PTS810SJK250SMTRLFS | 4 |
| TPS72325DBVR | IC REG LIN -2.5V 200MA SOT23-5 | U1 | SOT-23-5 | TPS72325DBVR | 1 |
| MCP1754T-3302E/OT | IC REG LINEAR 3.3V 150MA SOT23-5 | U2 | SOT-23-5 | MCP1754T-3302E/OT | 1 |
| TLV70025QDDCRQ1 | IC REG LINEAR 2.5V 200MA SOT23 | U3 | SOT-23-5 | TLV70025QDDCRQ1 | 1 |
| ADS1299IPAG | IC AFE 8 CHAN 24BIT 64TQFP | U4 | 64-TQFP (10x10) | ADS1299IPAG | 1 |
| PIC32MX250F128B-I/SS | IC MCU 32BIT 128KB FLASH 28SSOP | U5 | SSOP-28-208mil | PIC32MX250F128B-I/SS | 1 |
| LM2664M6/NOPB | IC REG CHRG PUMP INV 40MA SOT23 | U6 | SOT-23-6 | LM2664M6/NOPB | 1 |
| 8MHz | CRYSTAL 8.0000MHZ 18PF SMD | X1 | SMD6035-4P | ABMM2-8.000MHZ-E2-T | 1 |